

Train to Edinburgh - Grown-ups

UK Family Tour

One Island, Two Old Nations

Somewhere in the next hour or so, with no bridge to announce it and no guard to stamp anything, your train will slip out of one country and into another. It is worth pausing on what that actually means, because for a newcomer the geography of these islands can be quietly baffling. So here is the shape of it. You are riding up the spine of a single island, the one properly called Great Britain, and that one island has long held more than one nation. England fills the south and the centre, Scotland the north, and Wales the western bulge you cannot see from here. Three old countries on one island, with a scatter of smaller isles around them and, across the water, the separate island of Ireland.

England and Scotland are the two that concern you today, and they are genuinely two, not two halves of one thing. They were rival kingdoms for the better part of a thousand years, each with its own kings and its own armies, and a border between them that was fought over so often and so savagely that its scars are still on the map. They were joined, eventually, in stages. In sixteen oh three they came under a single crown, when a Scottish king inherited the English throne as well. Just over a century later, in seventeen oh seven, their two parliaments merged into one, and the United Kingdom as we know it began to take shape. And yet Scotland never dissolved into England. To this day it keeps its own laws, its own schools and universities, its own national church, even its own banknotes, and, fiercer than any of these, its own emphatic sense of itself. You are not changing counties this afternoon. You are crossing from one nation into another, on one island, under one crown, and the rest of your week belongs to the northern one.

Same Island, Still a Different Country

It is easy for a visitor to assume that England and Scotland long ago blended into a single British sameness. They did not, and the reasons are written into the very deal that joined them. Go back to sixteen oh three. Queen Elizabeth the First of England died childless, and the nearest heir happened to be her cousin's descendant, James, who was already King of Scots. So he rode south to London and wore both crowns at once, King James the Sixth of Scotland and First of England. This is what historians call the Union of the Crowns, and it is exactly that and no more, one man wearing two hats. The two countries still had their own parliaments, their own laws, their own everything, merely a shared monarch who mostly preferred the richer, warmer south.

The fuller union came in seventeen oh seven, and it was a harder, colder, more commercial affair, driven partly by a Scottish financial catastrophe and partly by English nerves about who would inherit the throne next. Under the Acts of Union the two parliaments dissolved themselves and reconvened as a single body, the Parliament of Great Britain, sitting at Westminster in London. That is the moment Great Britain became one state rather than two countries that happened to share a king.

But here is the clever, telling detail. The Scots who signed up in seventeen oh seven insisted on keeping the things that made them a nation, and they got it in writing. The treaty specifically guaranteed that Scotland would keep its own legal system, its own separate system of education, and its own national church, the Presbyterian Church of Scotland, quite distinct from the Church of England. Three centuries

later, those guarantees still hold. A Scottish lawyer trains in a different body of law from an English one. A Scottish child moves through a different school system toward different exams. And the pound in your pocket may soon look strange, because Scottish banks print their own banknotes, real money, perfectly valid, and utterly different in design from the notes you changed at the airport. English shopkeepers sometimes squint at them, and that little squint is three hundred years of history in miniature. So no, you have not simply crossed into the northern end of England. You have crossed into a country that agreed to share a state while carefully keeping its soul.

The Iron Road That Shrank the Island

Think for a moment about what this journey used to cost, not in money but in days. Before the railway came, getting from London to Edinburgh meant something close to a small expedition. The fastest mail coaches, changing horses every few miles and rattling on through the nights, took the better part of two days and left you thoroughly battered. For most people the two ends of this island were, in any practical sense, different worlds that rarely met. Then, in the space of a single decade in the eighteen forties, the railway arrived and quietly performed a kind of magic. It did not move the cities, but it moved them closer, folding a two-day ordeal down into a single comfortable day, and it did the same trick everywhere at once. An island that had always been a loose collection of regions, each with its own accent and its own horizon, suddenly became a place you could actually cross.

The trains changed something even stranger, something invisible. Before them, every town kept its own local time, set by its own noon sun, so a clock in the west might read several minutes behind a clock in the east, and nobody much minded, because nobody could travel fast enough for it to matter. The railways could not run on that. A timetable needs one time, the same time, everywhere along the line, so the companies imposed a single standard clock across the country, and ordinary Britain slowly gave up its patchwork of local noons for one national time. The railway did not just carry the country together. It synchronized it.

And so it is worth looking up from your book now and then and simply watching the land change, because the journey tells its own story. The soft, green, hedged lowlands of the south give way, mile by mile, to something barer and harder and older, more stone than hedge, more moor than meadow, the ground quietly rising toward the north. By the time the sea appears on your right and the hills gather on your left, the very look of the country is telling you what the border only confirms. You are somewhere new, and the soft south is well behind you.

The East Coast Main Line: Racing History

The four-hundred-mile London-to-Edinburgh run you're on is the East Coast Main Line, assembled in the 1840s and 50s by three companies and glamorized for a century by the Flying Scotsman — originally the name of the 10 AM departure from King's Cross, running daily since 1862, later lent to the most famous locomotive on Earth. The line's competitive history is genuinely entertaining: in the Races to the North of 1888 and 1895, East Coast and West Coast companies covertly raced overnight expresses to Scotland, quietly ignoring their own agreed timetables until a derailment scare ended the fun. The speed landmarks all happened on this route: Flying Scotsman's first authenticated hundred miles per hour in 1934, and Mallard's one hundred twenty-six in 1938 — still the world steam record — set on Stoke Bank south of Grantham, which you passed early in the journey. Sights worth timing your coffee for: Durham, roughly halfway, where the train curves on a viaduct giving a magnificent elevated view of the Norman cathedral and castle on their peninsula — right-hand side northbound, and widely rated among the finest railway views in Europe; then the Northumberland coast, where the line runs close to the North Sea past Alnmouth, with Lindisfarne — Holy Island — and Bamburgh Castle visible on clear days in the distance.

Related listening: The line races right past York; you'll walk its walls in a few days. Coming up — “From Eboracum to York: A Capital in Layers” (York & the Minster, Day 13).

Out of King's Cross: The Modern Racers and a Buried Queen

The train you're on continues a very old competition by thoroughly modern means. If this is one of the London North Eastern Railway expresses, it's an Azuma, the Hitachi fleet that swept away the old diesels in 2019. There's a pleasing loop in that. Azuma is a Japanese word meaning east, fitting for the East Coast line, yet the trains themselves are assembled in the north of England, at Hitachi's factory at Newton Aycliffe in County Durham, which sits just off this very route. Japanese engineering, County Durham labour, running the line the Flying Scotsman made famous, at around one hundred and twenty-five miles an hour without a wisp of smoke.

The departure itself is worth replaying. Out of King's Cross the train plunges almost at once into the Gasworks Tunnel, bored beneath the Regent's Canal, then begins the long climb that generations of drivers knew as the first real test of an engine. Glance left in the early minutes and you'll catch Alexandra Palace on its ridge, the vast Victorian hall from which the BBC transmitted the world's first regular high-definition television broadcasts in 1936. And while you're back at King's Cross in memory, spare a thought for its most photographed corner, the ironwork sign for Platform Nine and Three-Quarters, complete with a half-vanished luggage trolley, courtesy of a certain young wizard.

An hour or so out, the spires of Peterborough Cathedral rise from the flat Fenland to the right. It is one of the great Norman churches of Europe, and beneath its floor lies Catherine of Aragon, first queen of Henry the Eighth, buried here in 1536 after he cast her aside. For a quarter of a century it held a second queen as well. Mary, Queen of Scots was interred here after her execution at nearby Fotheringhay in 1587, until her son, by then King James of a united crown, had her moved to a grander tomb at Westminster. You pass the city of two queens at speed, on the fastest, flattest running anywhere on the line. Somewhere in that fast stretch you also slip past Grantham, an unshowy market town with two improbable sons. It is where the young Isaac Newton went to grammar school, a few miles from the orchard at Woolsthorpe where the apple is supposed to have fallen, and where, two and a half centuries later, Margaret Thatcher was born above her father's grocer's shop. The Fens hide their history well.

Gresley, the A4s, and Why Mallard Still Wins

Mallard's record deserves more than a number, because it was the summit of a specific and rather beautiful engineering moment. By the 1930s the railways were fighting the motor car and the airplane for glamour, and speed was the weapon of choice. The London and North Eastern Railway had a secret one, its chief engineer, Sir Nigel Gresley, and his A4 Pacifics. Gresley was unusually fascinated by aerodynamics for his age. He shaped the A4 with a smooth, wind-cheating profile said to owe something to the wedge nose of a Bugatti railcar, refined in wind-tunnel tests on scale models. The result was that sloping streamlined snout and the deep garter-blue livery, a machine that looked like the future and sliced through the air instead of fighting it.

He gave them evocative names, many of them birds, Mallard and Bittern and Kingfisher and Wild Swan among them. On the third of July, 1938, Mallard, barely four months old, was sent down Stoke Bank south of Grantham, ostensibly on a brake trial, driven by Joseph Duddington, a man known across the railway for his nerve. It touched one hundred and twenty-six miles an hour, snatching the record from a German locomotive that had managed one hundred and twenty-four and a half two years before. The effort overheated one of the big-end bearings, and the crew wisely eased off, but the number was in the book.

Here is the quietly astonishing part. That record has never been broken, and almost certainly never will be. Within twenty years the main-line railways were abandoning steam altogether for diesel and electric, so nobody seriously tried to build a faster steam engine again. Mallard hit a ceiling that turned out to be permanent, the high-water mark of an entire technology, frozen forever in July 1938. Of the thirty-five A4s that Gresley built, six survive, scattered as far as museums in the United States and Canada, and in 2013 all six were gathered nose to nose in York for the record's seventy-fifth anniversary, a reunion of the fastest steam locomotives the world will ever see. Mallard itself survives, immaculate in its blue, at the National Railway Museum in York, which your train passes. It is, in the most literal sense, the fastest that steam has ever gone, or ever will.

Prince-Bishops, the Tyne, and the Angel

The middle of the journey delivers its scenery in a rush. First Durham, roughly halfway, where the line swings onto a viaduct and hands you what many rate the finest railway view in Britain, the Norman cathedral and its castle rising together on a rock within a tight loop of the River Wear, on the right going north. Look quickly, because it is gone in seconds. That cathedral holds the bones of Saint Cuthbert and of the Venerable Bede, and for centuries Durham was ruled not by a mere bishop but by prince-bishops, churchmen granted near-royal powers, their own courts, their own coinage, even their own army, to hold England's wild northern frontier against the Scots. Cathedral and castle together were among Britain's very first UNESCO World Heritage Sites, listed in 1986. Within their palatinate the prince-bishops minted their own money and raised their own armies, sovereigns in miniature, and the King's writ effectively stopped at the county line, a peculiar arrangement that endured in name until the nineteenth century.

Minutes later comes Newcastle, and one of the great urban entrances by rail anywhere in the world. The train crosses the Tyne on the King Edward the Seventh Bridge of 1906, and for a few seconds the whole cluster of Tyne bridges lines up beside you downstream. The green steel arch of the Tyne Bridge, opened in 1928 and built by Dorman Long of Teesside, the same firm that raised the Sydney Harbour Bridge, which rather explains the family resemblance. Robert Stephenson's double-decked High Level Bridge of 1849, road slung below and rail above. And further down, the tilting Gateshead Millennium Bridge, the so-called Blinking Eye.

Then, on the rise near Gateshead, the sculpture appears, Antony Gormley's Angel of the North, unveiled in 1998, twenty metres tall with a wingspan of fifty-four metres, cast in weathering steel so it wears a permanent russet skin. Modelled on the artist's own body and planted where a colliery once stood, it is

seen by some thirty-three million people a year, most of them, like you, in passing. Gormley gave the wings a slight forward tilt, he said, to make an embrace rather than a threat. Consider yourself embraced, and note the theme of this whole region, which built the locomotives, forged the bridges, and now builds the very trains that carry you through it.

The Station You Left: Cubitt's Shed and a Buried Queen

Spare a backward glance for the building you walked out of, because King's Cross is one of the quiet masterpieces of Victorian London, and most people rush through it without ever looking up. It opened on the fourteenth of October, 1852, as the London terminus of the Great Northern Railway, and at the time it was the largest station in Britain. Its architect, Lewis Cubitt, did something almost radical for the age. While every other railway company was dressing its termini as cathedrals and Gothic palaces, Cubitt gave King's Cross two vast, plain, semicircular arches of honest yellow brick, echoing the twin train sheds behind them, topped by a single modest clock tower and almost no ornament at all. He is supposed to have boasted that the building would depend for its beauty on the fitness of its form for its purpose, and its size. It is the most functional great station in London, and next to the soaring pink fantasy of St Pancras, which the rival Midland Railway built beside it twenty years later almost as a taunt, it looks like a Quaker standing next to an opera singer.

For decades that clean facade was hidden. A cheap green passenger canopy was bolted across the front in the nineteen seventies and squatted there for a generation. Then came the great restoration for the twenty twelve Olympics, when engineers stripped the clutter away, revealed Cubitt's brick once more, and raised the extraordinary new Western Concourse alongside it, that white steel lattice fanning up and outward from a single central funnel like the vaulting of a palm tree or the underside of a vast web. It is, they will tell you, the largest single-span station structure in Europe, and it turned a grimy commuter shed back into a place worth standing still in.

And then there is the legend, which you should enjoy precisely as a legend. The story goes that Boudicca, the warrior queen of the Iceni who burned Roman London to the ground around the year sixty, lies buried somewhere beneath the platforms of King's Cross, with the favourite candidates being platforms eight, nine, and ten. There is essentially no evidence for it. The whole tale hangs on the fact that this district was once called Battle Bridge, which somebody, centuries later, decided must commemorate her last stand against the Romans. Historians roll their eyes. But savour the geography of it anyway, because if the myth were true, the rebel queen who defied an empire would be lying in eternal rest a few feet from platform nine and three quarters, where a certain luggage trolley disappears into the wall, and a queue of children waits to follow her.

The Racing Ground and a First Sight of York

For the first hour or two out of London the line does something it does almost nowhere else in Britain. It runs flat, straight, and fast, across the old kingdom of the Fens and then the broad Plain of York, and this is not an accident of geography so much as the whole point of the East Coast route. The engineers who drove it north chose these easy gradients deliberately, because a level line is a fast line, and speed was always this railway's reason to exist. This is the ground on which the record-breakers ran. The long, gentle falls and rises here let the great expresses stretch their legs, and it is why the timetable planners have always been able to promise London to Edinburgh in a shade over four hours. Watch the fields blur and understand that you are on the fastest classic main line in the country, purpose-built for exactly the sensation you are feeling.

Somewhere in the middle of that fast running the train slows and threads through York, and this is worth putting your book down for, because in a few days you will be back. On the thirteenth day of this trip you will spend proper time in York, walking its medieval walls and its huddled lanes, so treat this as the

trailer. As the train eases through the station, look out for the pale limestone towers of York Minster rising over the rooftops, the largest Gothic cathedral in Northern Europe, five centuries in the building and still, from certain angles, the only tall thing for miles. York was the Romans' great northern capital, the city where an emperor died and another was proclaimed, then a Viking kingdom in its own right, and you will meet that Viking city again too, at the JORVIK centre, where the same Norse world that fell on Lindisfarne in seventy nine three is dug up street by street. York station itself deserves a second's attention as you pass through it, because it is one of the loveliest railway buildings in Britain, a great sweeping curve of iron and glass built in the eighteen seventies, its roof following the bend of the tracks in three long arcs. When it opened it was briefly the largest station in the world. It is also, fittingly, home to the National Railway Museum, where Mallard sleeps in its blue livery and where, on the record's seventy-fifth anniversary, all six surviving streamlined A4s were gathered nose to nose. You will very likely walk in and see them on the thirteenth day.

For now you merely graze it all at speed. The Minster, a glimpse of the walls, the great curved roof, and gone. File it away. You will be walking those streets soon enough, and the train is already hurrying you north.

Berwick, the Border, and Arriving in Edinburgh

Berwick-upon-Tweed, the last English town, compresses the whole Anglo-Scottish story: it changed hands between the kingdoms around thirteen times before 1482, was for centuries so anomalous that it was named separately in royal proclamations — spawning the enduring myth that Berwick remained technically at war with Russia after the Crimean War — and its Elizabethan ramparts, built against a Scottish invasion that never came, were the most expensive undertaking of Elizabeth the First's reign. Your train crosses the Tweed on Robert Stephenson's Royal Border Bridge, twenty-eight arches completed in 1850, with the actual border a couple of miles further north at Marshall Meadows — marked only by signs and, on the adjacent A1, a piper on good days. The final approach to Edinburgh runs along the East Lothian coast, with the Firth of Forth opening on the right; in the distance you may catch the Bass Rock, a volcanic plug that hosts the world's largest colony of northern gannets and visibly whitens with birds in summer. Waverley station itself is a statement: the only station in Britain named after novels — Walter Scott's Waverley series — sunk in the valley between the medieval Old Town and the Georgian New Town, with the castle looming overhead as you exit. The geography lesson is immediate: everything in central Edinburgh is arranged around that extinct volcano, and tomorrow you'll walk its spine.

Related listening: Holy Island, seen from the train, is where the Viking Age began in 793. Coming up — “1066: The Viking Age Ends Just Up the Road” (JORVIK & Viking York, Day 14).

Lindisfarne and the Northumberland Coast

Before the border, the line runs close to a coast thick with history, so keep half an eye seaward. Watch east for Holy Island, Lindisfarne, a tidal island severed from the mainland twice a day when the sea floods its causeway. In the seventh century it was the cradle of Northumbrian Christianity, home to Saint Cuthbert and the workshop where the Lindisfarne Gospels were made, that miracle of illuminated manuscript now kept in the British Library. And it was here, in the year 793, that Viking raiders fell on the monastery in an attack so shocking to Christian Europe that many historians date the start of the Viking Age to that single day. Longships on this grey water, and the medieval world tilted on its axis.

The same coast is studded with fortresses, thanks to eight centuries of Anglo-Scottish violence that made every headland worth defending. The showpiece is Bamburgh Castle, an immense pile on a crag of black basalt above the dunes, seat of the Anglo-Saxon kings of Bernicia and in effect an early capital of the north. Inland stands Alnwick Castle, still the seat of the Dukes of Northumberland and familiar to any child as the Hogwarts of the early Harry Potter films. And on a lonely promontory rise the broken towers of Dunstanburgh, reachable only on foot and a favourite subject of the painter Turner. This is Whin Sill country, a great intrusion of hard volcanic rock that surfaces as cliffs and crags right along the shore, and the medieval builders exploited every outcrop of it.

Look to the water, too. Offshore lie the Farne Islands, seabird cities in summer and the place where Cuthbert lived out his last years as a hermit among the puffins and the grey seals. Those same rocks staged one of the great Victorian dramas. On a wild September night in 1838 a paddle-steamer broke apart on them, and a lighthouse keeper's daughter named Grace Darling rowed out through the storm with her father to haul the survivors from the sea, an exploit that made her the most famous woman in Britain almost overnight. She died young and lies in the churchyard at Bamburgh, her memorial deliberately raised high so that passing sailors could see it from the water. All told, it is one of the loveliest stretches of the entire east coast of Britain, and you are getting it for the price of a train ticket, gliding past at a mile a minute while everyone else has to walk.

The Border and the Debatable Land

The frontier you're crossing is one of the oldest and most fought-over in Europe, though today it slips by almost unmarked. The official line lies a couple of miles north of Berwick, near Marshall Meadows, and if you're watching the parallel road you may glimpse a Saltire and a Welcome to Scotland sign. The railway makes no fuss whatsoever. For three hundred years, though, this was among the most lawless corners of the continent, the country of the Border Reivers, raiding families on both sides of the line who spent the late Middle Ages stealing one another's cattle, burning one another's farms, and running protection rackets across the frontier. They were so accomplished at it that they left a word in the English language, blackmail, the illegal payment you handed over for the favour of being left in peace. The reiving families are scattered around the world now by the surnames they left behind, Armstrong and Elliot and Nixon and Kerr among them, and here is a fact worth carrying off the train: the first human being to set foot on the moon, Neil Armstrong, sprang from exactly this borderland stock. In 1974 he came in person to the little town of Langholm, just over the hills, to be made a freeman of the reivers' country, the raider's descendant returned from space.

Away to the west lay the Debatable Land, a strip that for centuries belonged officially to neither kingdom, because neither could govern it, a genuine no-man's-land where the ordinary rules simply lapsed. All of that ferocity ended in earnest only after 1603, when the crowns of England and Scotland were united under one king, James, and the border lost its purpose almost overnight.

Across the line the character of the run changes. The railway clings to the East Lothian coast, past the nuclear station at Torness and the fishing town of Dunbar, birthplace in 1838 of John Muir, the naturalist who did more than any other single person to create America's national parks, Yosemite chief among them. A Scot, in other words, who taught the New World to love its own wilderness, which is worth a quiet moment as an American family rolls past his front door. Then the outskirts of Edinburgh begin to gather, the land lifts, and the city's great volcanic set-pieces heave into view ahead of you.

Arthur's Seat, the Nor Loch, and Waverley

Understand Edinburgh's geology and you understand the entire city, because it is built on the wreckage of volcanoes and ice. The dramatic green mass rising to the east as you arrive is Arthur's Seat, the deeply eroded core of a volcano that last stirred more than three hundred million years ago. Castle Rock, the sheer black crag that carries Edinburgh Castle, is another ancient volcanic plug. During the ice ages, glaciers grinding from west to east slammed into that hard plug and were forced to part around it, scouring away the softer rock behind and leaving a long tapering ramp of debris in its lee. Geologists call the shape a crag and tail, and the medieval city simply obeyed it, the castle on the crag and the Royal Mile running down the tail. Tomorrow you will walk the anatomy of a glacier.

The station you arrive at occupies the strangest site of all. Waverley sits in the deep valley between the medieval Old Town and the Georgian New Town, but that valley was, until the eighteenth century, a body of water, the Nor Loch, a foul, sewage-fed lake that doubled as the town's rubbish dump and, conveniently, the place to drown accused witches. The Georgians drained it, and into the drained bed went the gardens, Princes Street, and eventually this vast station, the only one in the world named after a novel, Walter Scott's Waverley. Look up as you step onto the platform and you'll see Scott's own monument, a soaring Gothic rocket of soot-blackened stone, said to be the largest memorial to a writer anywhere on Earth, some two hundred feet of blackened Gothic pinnacle with sixty-four of Scott's own characters carved climbing its buttresses and a white marble Scott, with his deerhound at his knee, seated beneath.

And above it all, the castle on its rock, the reason a city grew here in the first place. You have travelled some four hundred miles up the spine of Britain, from the flatlands of the Fens to the bones of a volcano, in barely four and a half hours. Step down onto the platform, breathe the sharper northern air, and begin.

The Firth Opens and the Volcano Waits

Once you are over the border and running along the East Lothian coast, keep the seaward side, on your right, in view, because the water out there is about to change character. The open North Sea gradually gives way to a great silver arm of it reaching inland, and that is the Firth of Forth, the drowned mouth of the River Forth and the true threshold of Edinburgh. You are looking at your itinerary. In two days' time, on the tenth day of this trip, you will be out on that very water, sailing from South Queensferry to the little island of Inchcolm and its ruined medieval abbey, passing beneath the most famous railway bridge on Earth, the great red cantilever Forth Bridge of eighteen ninety, whose three vast steel diamonds have been shorthand for Victorian engineering nerve for well over a century. You will not quite see the bridge from this train, it lies off to the west, up the line that peels away toward Fife, but the water you are watching now is the water you will cross then. Consider the Firth an introduction.

Then, ahead and a little to the left, the land begins to heave up into something far too dramatic for an ordinary city edge. A long green ridge rears against the sky, culminating in a bare rocky summit, and that is Arthur's Seat, the eroded plug of a volcano that died more than three hundred million years ago and now sits, absurdly, in the middle of a capital city. On the twelfth day you will climb it, the whole family, up the grassy flank to that eight hundred and twenty foot crown and one of the great urban views in Europe, the city and the Firth and the hills of Fife all laid out beneath you. So when it slides into view now, do not just admire it. Size it up. That is your Tuesday. And beyond it, drawing closer with every mile, the black crag of the Castle Rock, another old volcano, with the fortress on its summit and the medieval town streaming down its tail toward the station where, in a few minutes, you will step off the train and into the trip proper.

Lumo, and the Race North Begins Again

Look around the carriage. There is no first class on this train, no dining car, no snowy tablecloths, just one honest standard-class cabin from end to end and a trolley rattling down the aisle. That is not a cost-cutting accident. It is the entire business model, and it makes the train you are riding one of the more interesting things on the British railway. You are on a Lumo service, and Lumo is something the East Coast Main Line had not seen in generations, a genuine competitor.

For most of modern history this line has been run by a single dominant operator, the state as often as not, presently the London North Eastern Railway, whose blue and red expresses you will have seen sweeping past. Lumo is a different animal. It is what the industry calls an open-access operator, a company that does not hold a government franchise but simply buys the right to run its own trains on the public tracks and takes its chances on filling the seats. It launched in October of twenty twenty one, owned by the transport group FirstGroup, with a fleet of just five sleek Japanese-designed trains, and a single, ruthless idea borrowed wholesale from the budget airlines. Strip out the frills, run one class, sell every seat cheap and sell it early online, and go head to head not really with the other railway at all, but with the aeroplane. The people Lumo most wants to beat are the ones flying from Heathrow and Gatwick up to Edinburgh, and its pitch to them is simple. City centre to city centre, no airport, no security queue, a fraction of the carbon, and often a fraction of the fare.

There is a serious card in that hand, too, and it is the reason a family with children might feel rather good about this particular train. Flying from London to Edinburgh and gliding there by electric train are not remotely equal when it comes to the carbon each passenger burns. A short domestic flight is one of the most wasteful ways to move a human being ever devised, and an electric train drawing clean power from the wire is one of the least. Lumo leans hard on exactly that, marketing itself as the low-carbon way north, and the sums are genuinely on its side. So there is a small, quiet virtue folded into your comfortable seat. You are getting there for less money, with more to look at, and leaving a far lighter footprint than the people in the vapour trails above you.

The trains themselves close a rather satisfying loop. This is a Class eight-oh-three, built by Hitachi, its bodysell shipped from Japan but its final assembly done in England, at Hitachi's factory at Newton Aycliffe in County Durham, which sits just off this very route and which you will pass this afternoon. It is all-electric, drawing power from the wire overhead, not a drop of diesel aboard, and it will cruise at a hundred and twenty-five miles an hour without a wisp of smoke. Japanese engineering, County Durham hands, running the line the Flying Scotsman made famous.

And here is the pleasing thing to hold in mind as the miles go by. You began this journey with the story of the Races to the North, when rival companies on the east and west coasts secretly flogged their overnight expresses toward Scotland in the eighteen nineties, each trying to shave minutes off the other, until the whole reckless game ended in an agreed truce. Competition on this line was then, for well over a century, effectively outlawed. What you are sitting in is its return. Lumo against the incumbent, budget against comfort, both of them racing the airlines overhead, and the winner, for once, is quite clearly the passenger. The old race north is on again, only this time it is fought with cheap advance fares instead of stopwatches, and you are riding the result.

How the Line Was Stitched Together

The four hundred smooth miles you are travelling feel like a single grand design, one clean line drawn from London to Edinburgh by some visionary hand. It was nothing of the sort. The East Coast Main Line is a patchwork, stitched together in a hurry in the eighteen forties by three separate and frequently squabbling companies, during the wildest financial frenzy Britain had yet seen, and it is a small miracle that the joins do not show.

That frenzy has a name, Railway Mania. In the middle of the eighteen forties the country went collectively mad for railway shares. Ordinary people who had never invested in anything poured their savings into hundreds of proposed lines, Parliament waved through act after act, and for a few giddy years railway schemes were floated faster than track could possibly be laid. Much of it was fantasy, and when the bubble burst, as bubbles do, thousands were ruined. But the mania also left behind, almost as a by-product, the skeleton of the national network Britain still runs on today, this line very much included. Presiding over the eastern portion of it was one of the great characters of the age, George Hudson, the so-called Railway King, a York draper turned tycoon who at his peak controlled a huge slice of Britain's railways and entertained on a royal scale. He was also, it turned out, running much of his empire as a fraud, paying dividends out of capital, and his eventual fall was as spectacular as his rise. The line you are on was built in his shadow, by the companies he schemed to control.

Three of them carried you north. The Great Northern brought the line up out of King's Cross across the flatlands; the North Eastern carried it on through Yorkshire and Durham and over the Tyne and the Tweed; and beyond the border the North British ran it in to Edinburgh. Each built its own great river crossings, which is why the Tyne is spanned by the Stephenson's work at Newcastle and the Tweed by Robert Stephenson's Royal Border Bridge, monuments to companies that no longer exist. It took until nineteen twenty three, and an Act of Parliament that forcibly merged Britain's railways into just four giants, for the whole route to fall under one owner at last, the London and North Eastern Railway, the legendary LNER of Gresley and Mallard and the streamlined age.

One inheritance from those early builders is quite literally beneath you, and it shaped railways across half the planet. When George Stephenson, Robert's father, was laying out his first lines in the coalfields of the north east, he set his rails four feet eight and a half inches apart, an oddly specific figure said to descend from the wheel ruts of horse-drawn colliery wagons, which in turn are sometimes traced, romantically and unreliably, all the way back to Roman cart tracks. Rival engineers argued for something broader and steadier, and for a while Britain fought a genuine gauge war, but Stephenson's narrower measure had been laid down first and laid down widest, and in eighteen forty six Parliament simply made it the law. That is why this train, and most trains from here to Australia, still run on rails set precisely four feet eight and a half inches apart, a decision taken by a self-taught engineer nearly two centuries ago and never seriously revisited since.

One last quirk to notice as another train flashes past your window. In Britain the trains keep to the left, just as the road traffic does, the up line and the down line running like a dual carriageway of steel. Much of the world drives its trains on the right, but Britain, which invented the main-line railway, naturally did it its own way and never changed, and the two expresses passing each other at a closing speed of two hundred and fifty miles an hour do so, very properly, on the left.